

2010.14265: equation evidence

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Image
2010.14265_E00012	12	■ 0.492	$\begin{aligned} & \pi^{\prime}(1) \rightarrow \pi^{\prime}(2): Z \perp_P W \mid \emptyset \ \& \ \pi^{\prime}(1) \rightarrow \pi^{\prime}(3): Z \perp_P Y \mid \{W\} \\ & Y \mid W \ \& \ \pi^{\prime}(1) \rightarrow \pi^{\prime}(4): Z \not\perp_P X \mid \{W, Y\} \\ & \pi^{\prime}(2) \rightarrow \pi^{\prime}(3): W \not\perp_P Y \mid \{Z\} \\ & \pi^{\prime}(2) \rightarrow \pi^{\prime}(4): W \not\perp_P X \mid \{Z, Y\} \\ & \pi^{\prime}(3) \rightarrow \pi^{\prime}(4): Y \not\perp_P X \mid \{Z, W\} \end{aligned}$	$\begin{aligned} & \pi^{\prime}(1) \rightarrow \pi^{\prime}(2): Z \perp_P W \mid \emptyset \\ & \pi^{\prime}(1) \rightarrow \pi^{\prime}(3): Z \perp_P Y \mid \{W\} \\ & \pi^{\prime}(1) \rightarrow \pi^{\prime}(4): Z \not\perp_P X \mid \{W, Y\} \\ & \pi^{\prime}(2) \rightarrow \pi^{\prime}(3): W \not\perp_P Y \mid \{Z\} \\ & \pi^{\prime}(2) \rightarrow \pi^{\prime}(4): W \not\perp_P X \mid \{Z, Y\} \\ & \pi^{\prime}(3) \rightarrow \pi^{\prime}(4): Y \not\perp_P X \mid \{Z, W\} \end{aligned}$	$\begin{aligned} & \pi^{\prime}(1) \rightarrow \pi^{\prime}(2): Z \perp_P W \mid \emptyset \\ & \pi^{\prime}(1) \rightarrow \pi^{\prime}(3): Z \perp_P Y \mid \{W\} \\ & \pi^{\prime}(1) \rightarrow \pi^{\prime}(4): Z \not\perp_P X \mid \{W, Y\} \\ & \pi^{\prime}(2) \rightarrow \pi^{\prime}(3): W \not\perp_P Y \mid \{Z\} \\ & \pi^{\prime}(2) \rightarrow \pi^{\prime}(4): W \not\perp_P X \mid \{Z, Y\} \\ & \pi^{\prime}(3) \rightarrow \pi^{\prime}(4): Y \not\perp_P X \mid \{Z, W\} \end{aligned}$
2010.14265_E00009	11	■ 0.578	$\begin{aligned} & \pi(1) \rightarrow \pi(4): X \not\perp_P Y \mid \{Z, W\} \\ & \pi(2) \rightarrow \pi(4): Z \not\perp_P Y \mid \{X, W\} \\ & \pi(3) \rightarrow \pi(4): W \perp_P Y \mid \{X, Z\} \end{aligned}$	$\begin{aligned} & \pi(1) \rightarrow \pi(4): X \not\perp_P Y \mid \{Z, W\} \\ & \pi(2) \rightarrow \pi(4): Z \not\perp_P Y \mid \{X, W\} \\ & \pi(3) \rightarrow \pi(4): W \perp_P Y \mid \{X, Z\} \end{aligned}$	$\begin{aligned} & \pi(1) \rightarrow \pi(4): X \not\perp_P Y \mid \{Z, W\} \\ & \pi(2) \rightarrow \pi(4): Z \not\perp_P Y \mid \{X, W\} \\ & \pi(3) \rightarrow \pi(4): W \not\perp_P Y \mid \{X, Z\} \end{aligned}$
2010.14265_E00008	11	■ 0.675	$X_{\{\pi(j)\}} \not\perp_P X_{\{\pi(k)\}} \mid \{X_{\{\pi(1)\}}, X_{\{\pi(2)\}}, \dots, X_{\{\pi(k-1)\}}\} \setminus \{X_{\{\pi(j)\}}\},$	$X_{\pi(j)} \not\perp_P X_{\pi(k)} \mid \{X_{\pi(1)}, X_{\pi(2)}, \dots, X_{\pi(k-1)}\} \setminus \{X_{\pi(j)}\},$	$X_{\pi(j)} \not\perp_P X_{\pi(k)} \mid \{X_{\pi(1)}, X_{\pi(2)}, \dots, X_{\pi(k-1)}\} \setminus \{X_{\pi(j)}\},$
2010.14265_E00013	13	■ 0.706	$Z \perp_{\mathcal{G}} \{X, Y\} \mid \text{Pa}(Z).$	$Z \perp_G \{X, Y\} \mid \text{Pa}(Z).$	$Z \perp_G \{X, Y\} \mid \text{Pa}(Z).$
2010.14265_E00001	4	■ 0.823	$\forall S \subseteq V \setminus \{X, Y, Z\}: A \not\perp_P B \mid S \cup \{X, Y, Z\} \setminus \{A, B\},$	$\forall S \subseteq V \setminus \{X, Y, Z\}: A \not\perp_P B \mid S \cup \{X, Y, Z\} \setminus \{A, B\},$	$\forall S \subseteq V \setminus \{X, Y, Z\}: A \not\perp_P B \mid S \cup \{X, Y, Z\} \setminus \{A, B\},$
2010.14265_E00006	11	■ 0.874	$\begin{aligned} & P(X=1, \& W=1, Y=1, Z=1)=\frac{pq}{4}, \ \& \ P(X=1, Y=1, Z=1)=\frac{q}{4}, \ \& \ P(W=1, Y=1, Z=1)=\frac{p}{4}. \end{aligned}$	$\begin{aligned} P(X=1, W=1, Y=1, Z=1) &= \frac{pq}{4}, \\ P(X=1, Y=1, Z=1) &= \frac{q}{4}, \\ P(W=1, Y=1, Z=1) &= \frac{p}{4}. \end{aligned}$	$\begin{aligned} P(X=1, W=1, Y=1, Z=1) &= \frac{pq}{4}, \\ P(X=1, Y=1, Z=1) &= \frac{q}{4}, \\ P(W=1, Y=1, Z=1) &= \frac{p}{4}. \end{aligned}$
2010.14265_E00015	13	■ 0.960	$X \perp_{\mathcal{G}} Z \mid \text{Pa}(X) \cup (X \setminus \{X\}) \cup (Z \setminus \{Z\}),$	$X \perp_G Z \mid \text{Pa}(X) \cup (X \setminus \{X\}) \cup (Z \setminus \{Z\}),$	$X \perp_G Z \mid \text{Pa}(X) \cup (X \setminus \{X\}) \cup (Z \setminus \{Z\}),$
2010.14265_E00014 (1)	13	■ 0.980	$X \perp_{\mathcal{G}} Z \mid \text{Pa}(X) \cup (X \setminus \{X\}) \cup (Z \setminus \{Z\}),$	$X \perp_G Z \mid \text{Pa}(X) \cup (X \setminus \{X\}) \cup (Z \setminus \{Z\}),$	$X \perp_G Z \mid \text{Pa}(X) \cup (X \setminus \{X\}) \cup (Z \setminus \{Z\}),$

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2010.14265_EQ0011	12	■ 0.998	$\begin{aligned} & \pi^{\prime}(1) \rightarrow \pi^{\prime}(3): X \not\perp_P Y \mid \{Z\} \\ & \pi^{\prime}(2) \rightarrow \pi^{\prime}(3): Z \not\perp_P Y \mid \{X\} \\ & \pi^{\prime}(3) \rightarrow \pi^{\prime}(4): Y \not\perp_P W \mid \{X, Z\} \\ & \pi^{\prime}(1) \rightarrow \pi^{\prime}(4): X \not\perp_P W \mid \{Z, Y\} \\ & \pi^{\prime}(2) \rightarrow \pi^{\prime}(4): Z \not\perp_P W \mid \{X, Y\} \end{aligned}$	$\begin{aligned} & \pi^{\prime}(1) \rightarrow \pi^{\prime}(3): X \not\perp_P Y \mid \{Z\} \\ & \pi^{\prime}(2) \rightarrow \pi^{\prime}(3): Z \not\perp_P Y \mid \{X\} \\ & \pi^{\prime}(3) \rightarrow \pi^{\prime}(4): Y \not\perp_P W \mid \{X, Z\} \\ & \pi^{\prime}(1) \rightarrow \pi^{\prime}(4): X \not\perp_P W \mid \{Z, Y\} \\ & \pi^{\prime}(2) \rightarrow \pi^{\prime}(4): Z \not\perp_P W \mid \{X, Y\} \end{aligned}$	$\begin{aligned} & \pi^{\prime}(1) \rightarrow \pi^{\prime}(3): X \not\perp_P Y \mid \{Z\} \\ & \pi^{\prime}(2) \rightarrow \pi^{\prime}(3): Z \not\perp_P Y \mid \{X\} \\ & \pi^{\prime}(3) \rightarrow \pi^{\prime}(4): Y \not\perp_P W \mid \{X, Z\} \\ & \pi^{\prime}(1) \rightarrow \pi^{\prime}(4): X \not\perp_P W \mid \{Z, Y\} \\ & \pi^{\prime}(2) \rightarrow \pi^{\prime}(4): Z \not\perp_P W \mid \{X, Y\} \end{aligned}$
2010.14265_EQ0010	12	■ 1.000	$\begin{aligned} & \pi(1) \rightarrow \pi(2): X \perp_P Z \mid \emptyset \\ & \pi(1) \rightarrow \pi(3): X \perp_P W \mid Z \\ & \pi(2) \rightarrow \pi(3): Z \perp_P W \mid X. \end{aligned}$	$\begin{aligned} & \pi(1) \rightarrow \pi(2): X \perp_P Z \mid \emptyset \\ & \pi(1) \rightarrow \pi(3): X \perp_P W \mid Z \\ & \pi(2) \rightarrow \pi(3): Z \perp_P W \mid X. \end{aligned}$	$\begin{aligned} & \pi(1) \rightarrow \pi(2): X \perp_P Z \mid \emptyset \\ & \pi(1) \rightarrow \pi(3): X \perp_P W \mid Z \\ & \pi(2) \rightarrow \pi(3): Z \perp_P W \mid X. \end{aligned}$
2010.14265_EQ0016	13	■ 1.000	$X \perp_P Z \mid \boldsymbol{S} \cup \{Y\} \cup (\boldsymbol{X} \setminus \{X\}) \cup (\boldsymbol{Z} \setminus \{Z\}).$	$X \perp_P Z \mid \boldsymbol{S} \cup \{Y\} \cup (\boldsymbol{X} \setminus \{X\}) \cup (\boldsymbol{Z} \setminus \{Z\}).$	$X \perp_P Z \mid \boldsymbol{S} \cup \{Y\} \cup (\boldsymbol{X} \setminus \{X\}) \cup (\boldsymbol{Z} \setminus \{Z\}).$
2010.14265_EQ0002	6	■ 1.000	$Y := ((X \oplus Z) \wedge W) \oplus E,$	$Y := ((X \oplus Z) \wedge W) \oplus E,$	$Y := ((X \oplus Z) \wedge W) \oplus E,$
2010.14265_EQ0003	11	■ 1.000	$Y := ((X \oplus Z) \wedge W) \oplus E.$	$Y := ((X \oplus Z) \wedge W) \oplus E.$	$Y := ((X \oplus Z) \wedge W) \oplus E.$
2010.14265_EQ0004	11	■ 1.000	$\begin{aligned} & P(X=0, Z=1, Y=1) = P(X=0, Z=1) \cdot P(Y=1) \\ & \Leftrightarrow p + q - 2pq = q + \frac{p}{2} - pq \\ & \Leftrightarrow p - pq = \frac{p}{2}. \end{aligned}$	$\begin{aligned} & P(X=0, Z=1, Y=1) = P(X=0, Z=1) \cdot P(Y=1) \\ & \Leftrightarrow p + q - 2pq = q + \frac{p}{2} - pq \\ & \Leftrightarrow p - pq = \frac{p}{2}. \end{aligned}$	$\begin{aligned} & P(X=0, Z=1, Y=1) = P(X=0, Z=1) \cdot P(Y=1) \\ & \Leftrightarrow p + q - 2pq = q + \frac{p}{2} - pq \\ & \Leftrightarrow p - pq = \frac{p}{2}. \end{aligned}$
2010.14265_EQ0005	11	■ 1.000	$P(X, W \mid Y, Z) = \frac{P(X, W, Y, Z)}{P(Y, Z)}.$	$P(X, W \mid Y, Z) = \frac{P(X, W, Y, Z)}{P(Y, Z)}.$	$P(X, W \mid Y, Z) = \frac{P(X, W, Y, Z)}{P(Y, Z)}.$
2010.14265_EQ0007	11	■ 1.000	$pq = \frac{pq}{2P(Y=1)}.$	$pq = \frac{pq}{2P(Y=1)}.$	$pq = \frac{pq}{2P(Y=1)}.$